

M003 Week 7 — English Worksheet (Beginner)

Topic: Voxel Statue — Building Layer by Layer · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This week you build a statue **one layer at a time**. Each layer is a small grid, and the layers stack **upward along y**. In the plans below, means stone and means air.

1 · Predict 🧠

Read the steps. Before you imagine it happening, circle or write your answer.

```
layer 1:  # # #      layer 2:  . # .
          # # #          . # .
          # # #          . # .

build layer 1 at height 0
build layer 2 at height 1
```

Layer 1 is big and layer 2 is small. Does the statue get thinner at the top? Circle one: yes · no

Why?


```
build layer 2 at height 0
build layer 1 at height 1
```

Same layers, swapped order. Is the big layer on top now? Circle one: yes · no

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The layers should go in order: 1 at the bottom, 3 at the top.

```
# clean
build layer 1 at height 0
build layer 2 at height 1
build layer 3 at height 2
```

```
# buggy
build layer 3 at height 0
build layer 2 at height 1
build layer 1 at height 2
```

What is wrong? What does the buggy statue look like?

Pair B — Every layer should start at the **same corner** so the statue stands straight.

```
# clean
build layer 1 at (0, 0, 0)
build layer 2 at (0, 1, 0)
```

```
# buggy
build layer 1 at (0, 0, 0)
build layer 2 at (1, 1, 0)
```

What is wrong?

3 · Fill the Gap

Each layer goes **one higher** than the last. One number is missing. Fill it in using the word bank.

```
build layer 1 at height 0  
build layer 2 at height ____  
build layer 3 at height 2
```

Word bank: 1 · 5 · 0

Write the missing number:

4 · Tell Me What You Built 📷

Now switch to your homework world. Draw a small **layer plan** — 3 layers — then build your statue layer by layer, going up one height at a time. When you finish, come back here.

Send a photo or video of your statue, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My statue has ... layers ...

Each layer went one higher because ...

One tricky moment was when ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while you show your statue in the world. Talk like you are teaching a friend. Try to use these words: **layer, height, stack, plan.**

1. Show your layer plan on paper, then show the finished statue.
2. Point at the bottom layer and the top layer and say their heights.
3. Say in your own words why the order of the layers matters.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.